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HyperFree: A Channel-adaptive and Tuning-free Foundation Model for Hyperspectral Remote Sensing Imagery

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Abstract

Advanced interpretation of hyperspectral remote sensing images benefits many precise Earth observation tasks. Recently, visual foundation models have promoted the remote sensing interpretation but concentrating on RGB and multispectral images. Due to the varied hyperspectral channels, existing foundation models would face image-by-image tuning situation, imposing great pressure on hardware and time resources. In this paper, we propose a tuning-free hyperspectral foundation model called HyperFree, by adapting the existing visual prompt engineering. To process varied channel numbers, we design a learned weight dictionary covering full-spectrum from $0.4 \sim 2.5\,\mu m$, supporting to build the embedding layer dynamically. To make the prompt design more tractable, HyperFree can generate multiple semantic-aware masks for one prompt by treating feature distance as semantic-similarity. After pre-training HyperFree on constructed large-scale high-resolution hyperspectral images, HyperFree (1 prompt) has shown comparable results with specialized models (5 shots) on 5 tasks and 11 datasets. Code and dataset are accessible at https://rsidea.whu.edu.cn/hyperfree.htm.

1. Introduction

Hyperspectral imaging has been recognized by the European Union as one of the Top 100 Disruptive Technologies [58], due to its ability to provide high-resolution spectral data for precise Earth observation across diverse applications [64], including environmental monitoring [31], agriculture [69], and defense [20]. Unlike traditional RGB or multispectral imagery, Hyperspectral imagery (HSI) offers several unique features: high spectral resolution (typically 10 nm), an extensive wavelength range (400–2500 nm), and dozens to hundreds of spectral channels [64], which enable the detailed identification of diverse targets, while also increasing the complexity of data processing and analysis.

Recently, remote sensing foundation models (RSFMs) have demonstrated advanced interpretive capabilities, primarily focusing on multispectral and RGB images [4, 7, 13, 25, 38, 49], while also opening new possibilities for analyzing hyperspectral images. RSFMs follow two main

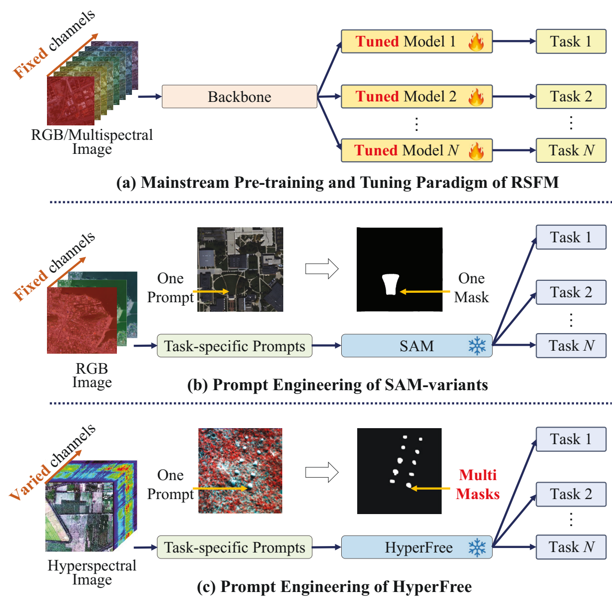

Figure 1. Comparison of existing RSFMs with the proposed HyperFree model. (a) The pre-training and tuning paradigm is the mainstream approach, where fine-tuning is essential for handling unseen HSIs. (b) The prompt engineering paradigm reduces the tuning burden but struggles to handle HSIs with hundreds of spectral channels. (c) HyperFree can directly process unseen HSIs with varying channel counts without fine-tuning, and it can identify multiple masks of the same class using only a single prompt.

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paradigms. The predominant paradigm is pre-training and tuning (P-T), illustrated in Figure 1(a), which begins with training a large-scale backbone model in a self-supervised manner, such as masked autoencoders (MAE) [12], followed by fine-tuning with task-specific heads for various downstream applications [7, 13, 38, 49]. In the P-T paradigm, both the pre-training and the fine-tuning stages can be time-consuming. Second paradigm is prompt engineering (P-E), as illustrated in Figure 1(b), which focuses on designing task-specific prompts for a frozen Segment Anything Model (SAM) [17] to directly perform downstream tasks without additional model tuning [4, 25], making it more efficient and user-friendly in practical.

The P-E paradigm is particularly well-suited for processing HSI, as it can be directly applied to diverse, unseen scenes without requiring model fine-tuning. However, it faces two significant challenges when applied to HSI. (i) The first challenge is the variation in spectral parameters across different hyperspectral sensors, including differences in spectral resolution, wavelength range, and the number of spectral channels [10]. For a P-T-based model like HyperSigma [54], although it is pre-trained using HSIs with fixed channels, the challenge can be addressed during its tuning stage by rebuilding the channel embedding layer to align the dimensions with the new data. But for a P-E-based model, the challenge of channel variation requires more careful consideration. (ii) The second challenge is to make prompt input tractable, ensuring a P-E based model can directly process unseen HSIs with least prior knowledge. Specifically, most existing P-E-based RSFMs rely on a frozen SAM [17], which segments remote sensing objects based on specialized models to generate prompts [4, 25]. As shown in Figure 1(b), each prompt typically generates a single mask and all the prompts for target objects are needed by existing P-E-based RSFMs, which largely reduced the model feasibility.

To address the above challenges, as shown in Figure 1 (c), we propose *a channel-adaptive, tuning-free hyperspectral foundation model, called HyperFree*, which can process any unseen HSIs directly in a P-E manner. Specifically, (i) Motivated by the word dictionary in natural language processing, which stores vector embeddings as values corresponding to words and phrases of varying lengths as keys [32], HyperFree introduces a learnable wavelength weight dictionary in the embedding layer, spanning the 400 to 2500 nm spectral range in 10 nm intervals. This allows for the dynamic encoding of unseen HSIs with varying spectral parameters into visual tokens, all with consistent dimensionality. (ii) To make the prompt of HyperFree more tractable, HyperFree is designed to generate multiple semantic-aware masks for one prompt as in Figure 1 (c), where HyperFree maps each prompt and single mask from image space into feature space and treats the feature distance as semantic-similarity. Prompt, mask and feature are interacted adaptively for different downstream task in tuning-free manner with only one prompt or zero shot.

Due to the high acquisition cost and label difficulty, current hyperspectral datasets only have a few images [45, 69], far from the requirements to train the promptable HyperFree. To conquer this problem, we developed the Hyper-Seg data engine, in addition to utilizing well-known multispectral datasets (e.g., fMoW [5] and SpaceNet [50, 51, 59]). This engine automatically generates a large-scale annotated HSI dataset, consisting of nearly 50,000 pairs of HSIs and their corresponding segmentation masks. The original data are sourced from AVIRIS airborne hyperspectral sensors, with a spatial size of 512×512 pixels and 224 spectral channels spanning from 400 to 2500 nm at a 10 nm spectral resolution. Unlike other RSFMs trained on unannotated satellite hyperspectral imagery with a 30m spatial resolution [54], the Hyper-Seg data engine features higher spatial resolution (0.6 to 5.0 m), along with annotations of 15.44 million masks. It is currently the largest known hyperspectral dataset with high spatial resolution.

To demonstrate the model capabilities, we evaluated HyperFree directly in a P-E manner on 11 datasets across 5 segmentation-related tasks, including multi-class classification [21], one-class classification [66], target detection [65], anomaly detection [47], and change detection [26]. Additionally, we verified HyperFree in a P-T manner on 14 datasets spanning 8 tasks, with additional tasks including hyperspectral denoising [27], hyperspectral unmixing [37], and hyperspectral object tracking [61]. The experimental results show that HyperFree achieves comparable accuracy to state-of-the-art models in a tuning-free manner (using just one prompt), and outperforms most models after tuning.

Our contributions can be summarized as:

- We propose the first tuning-free hyperspectral foundation model, which can process any hyperspectral image in different tasks with promptable or zero-shot manner.
- A weight dictionary that spans the full spectrum, enabling dynamic generation of the embedding layer dynamically according to input wavelengths.
- We propose to map both prompts and masks into feature space to identify multiple semantic-aware masks for one prompt, where different interaction workflows are designed for each downstream tasks.
- We built the Hyper-Seg data engine to train the HyperFree model and tested it on 11 datasets from 5 tasks in tuning-free manner, 14 datasets from 8 tasks in tuning manner as an extensive experiment.

2. Related Work

Hyperspectral image processing. There are diverse tasks to process such precise observation with different targets

[31]. (a) Classification: Assigning a specific class to each pixel [21]. (b) One class classification: Discriminating the pixels of target class with labeled binary map [33]. (c) Target detection. Discriminating the pixels of target class with labeled spectrum. (d) Anomaly detection. Discriminating the pixels spectrally deviating from the background [24]. (e) Change detection: Discriminating the changing pixels between two time-steps [26]. Since the output of the above tasks is all pixel-level segmentation maps, it is feasible to unify them using the proposed HyperFree model in tuning-free manner.

Foundation models. Powered by the transformer architectures, foundation models are characterized by large-scale property in model size and pre-training data, and the transferring ability for downstream tasks [60]. In remote sensing community, some foundation models for multi-spectral images have already been proposed such as SatMAE [7], SepctralEarth [1] and SpectralGPT [13], which trains large ViT-like models [9] with the MAE strategy [12] and massive open-source multi-spectral images (13 channels). In contrast, hyperspectral images have a high acquiring cost [56] and make the large-scale pre-training stage more difficult. A recent study broke the situation by building a large-scale hyperspectral dataset HyperGlobal-450K and trained a foundation model called HyperSigma with MAE strategy as well [54]. Our work differs from HyperSigma in two aspects: (i) HyperSigma accepts fixed number of image channels, while HyperFree can deal with any number of channels directly. (ii) HyperSigma needs to be fine-tuned for each downstream task. Differently, HyperFree has ability to process different tasks in fine-tuning free manner.

Prompt engineering (P-E). Different from the pre-training and then fine-tuning paradigm, P-E can utilize pre-trained foundation models to complete downstream tasks directly with prompts [34]. For pre-trained large language models, prompts are mostly hand-crafted text to guide the model generate response for specific task [34, 42]. In computer vison community, SAM follows the same spirit and designs diverse prompts (e.g., point and box) to achieve few-shot transferring performance [17]. Our work extends the visual prompt engineering in hyperspectral data with varying image channels and supports five tasks in tuning-free manner with designed prompt-mask-feature interaction.

3. Hyper-Seg Data Engine

Hyperspectral image processing always refers to various dense tasks and a large-scale segmentation dataset is needed to apply the P-E. However, most hyperspectral segmentation datasets only have a few images due to the high acquisition and label cost, which is quite different from the million-scale dataset with natural images such as SA-1B [17]. To tackle this, we built a data engine called Hyper-Seg to generate segmented masks automatically for spectral images and expand the data scale. Hyper-Seg engine firstly separates the spectral image into several groups with three channels according to the prior of key channels position. Each group is then segmented by the SAM-H model [17] and the results are combined by NMS operation to eliminate redundance and output the final segmentation map. Figure 2 shows the overall workflow, where the nine key channels are selected according to the expert knowledge as in Supp.1.1. Considering the high spatial resolution property of airborne platform, we collected 41946 hyperspectral images from AVIRIS airborne sensor and applied the Hyper-Seg Engine. Besides, we also processed two existing multispectral datasets (fMoW [5] and SpaceNet [50, 51, 59]) to increase the scale further. Compared to the traditional hyperspectral annotation methods which refer to high-resolution RGB imagery, our data engine can utilize more spectral information and obtain a high-precision segmentation mask without labor cost.

Table 1 shows the comparable statistical information of training datasets between existing RSFMs and proposed HyperFree. Most models only concentrate on the RGB and multispectral images with channel number 3 13. Although the recent model HyperSigma [54] has collected a large-scale hyperspectral images, the spatial resolution (30m/pixel) is largely lower than us (0.6 5m/pixel), and the dataset scale (considering image size and number together) is relatively small. With the designed Hyper-Seg data engine, constructed dataset is the only one with segmentation masks supporting the promptable training. More detailed statistical information of our generated dataset is given in Supp.1.2. The final generated dataset has nearly 150k images and 15.44milion masks. We name both the data engine and generated dataset as the Hyper-Seg for simplicity.

4. HyperFree Foundation Model

HyperFree has two customed components upon the meta-architecture of SAM including channel-adaptive embedding and prompt-mask-feature (PMF) interaction as in Figure 3. Channel-adaptive embedding constructs the embedding layer dynamically according to varying input wavelength, and outputs tokens in same representation space. After the promptable training stage, PMF interaction discriminates the semantic-aware masks for downstream tasks by mapping both prompt and mask into feature space and treating feature distance as semantic similarity.

4.1. Channel-adaptive Embedding

To make HyperFree process any hyperspectral image, channel-adaptive embedding layer is firstly designed to map input image with varying channels into tokens with fixed length, inspired by the tokenizer in NLP community [32].

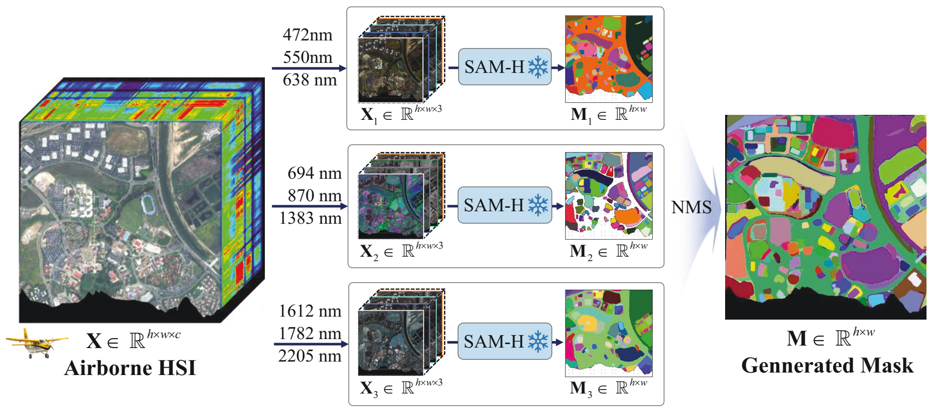


Figure 2. Hyper-Seg Data Engine automatically generates a large-scale dataset comprising nearly 41,900 pairs of airborne HSIs and their corresponding segmentation masks, enabling HyperFree to effectively learn discriminative spectral features.

Table 1. Comparison of training data used in current RSFMs. The primary features of the proposed HyperFree model include prompt engineering and the use of high-resolution annotated hyperspectral (HRS) imagery.

Dataset	Foundation Model	Spatial Resolution (m)	Spectral Resolution (nm)	Channel Number	Image Size	Image Number	Main Modality	Training Paradigm
RGB	RVSA [53]	$0.5 \sim 153$	\	3	$110 \sim 31672$	1.00×10^6	H/M/LSR-RGB	P-T
	RingMo [48]	$0.1 \sim 30$	\	3	448	2.10×10^6	H/MSR-RGB	P-T
	ScaleMAE [39]	$0.31 \sim 3.7$	\	3	\	3.64×10^6	HSR-RGB	P-T
Multispectral	SatMAE [6]	10	$15 \sim 180$	13	$\approx 45 \times 60$	7.13×10^5	MSR-M	P-T
	SkySense [11]	10/20	$15 \sim 180$	10	\	\	MSR-M	P-T
	SpectralGPT [14]	10/20/60	$15 \sim 180$	12	$45 \sim 120$	1.47×10^6	MSR-M	P-T
Hyperspectral	HyperSigma [54]	30	5/6.5/10	150/175	64	4.47×10^5	LSR-H	P-T
	HyperFree	$0.6 \sim 5.0$	10	224	512	4.19×10^4	HSR-H	P-E

* P-T: Pretraining and Tuning; P-E: Prompt Engineering.
* HSR: High Spatial Resolution; MSR: Medium Spatial Resolution; LSR: Low Spatial Resolution.
* RGB: Red-Green-Blue Imagery; M: Multispectral Imagery; H: Hyperspectral Imagery

Tokenizers rely on a pre-defined dictionary to break down a string of text into numerical tokens, where the dictionary has stored the numerical vectors of varied-length words or phrases. Vision transformers always use a convolutional layer to generate patch tokens and *what if we construct a layer weight dictionary treating each wavelength as a word*? Given the input meta data of wavelengths, we can just loop up the corresponding weights and construct the layer dynamically without re-training it.

The weight dictionary is built depending on the wavelengths to be processed and the expected token dimension. Since hyperspectral imaging mostly captures the spectral signals from 400nm to 2500nm and the 10nm is of high spectral resolution currently, 221 index channels are designed as the dictionary keys. Denote the output token as a vector of $\mathbb{R}^j$ corresponding to the patch size $p \times p$, each index channel stores a weight matrix of size $\mathbb{R}^{p\times p\times j}$, and the built dictionary β has a size of $221 \times p \times p \times j$. Given the input hyperspectral image $\mathbb{R}^{h\times w\times n}$ with wavelengths $\mathbf{b} = [b_1, b_2, \ldots, b_n]$, the corresponding kernel weight can be dynamically generated by combining the searched weight $\mathbf{W}$ with size $n \times p \times p \times j$, where n is flexible for each image. For conciseness, we use function g to represent the mapping process from $\mathbf{b}$ to $\mathbf{W}$ as kernel weight, and function f to represent the token generation step as in Equation (1).

$$f(\mathbf{X} \mid \mathbf{b}, \beta) = \text{Conv}_{\mathbf{W}}(\mathbf{X}) \quad \text{where} \quad \mathbf{W} = g(\mathbf{b}, \beta) \qquad (1)$$

In the past few decades, many satellites have been launched equipped with spectral sensors, where the key channels are selected by professional teams and practical experience. To utilize these successful priors, we further extract visual tokens in two parallel branches with dynamically constructed layers: one to process key channels and the other to process the split cubes, as shown in Figure 3. The input image $\mathbf{X}$ is first divided into key channels $\mathbf{X_k}$ and the cubes $\mathbf{X_c}$ between them, where the key wavelengths are selected by referring to famous spectral sensors (detailed

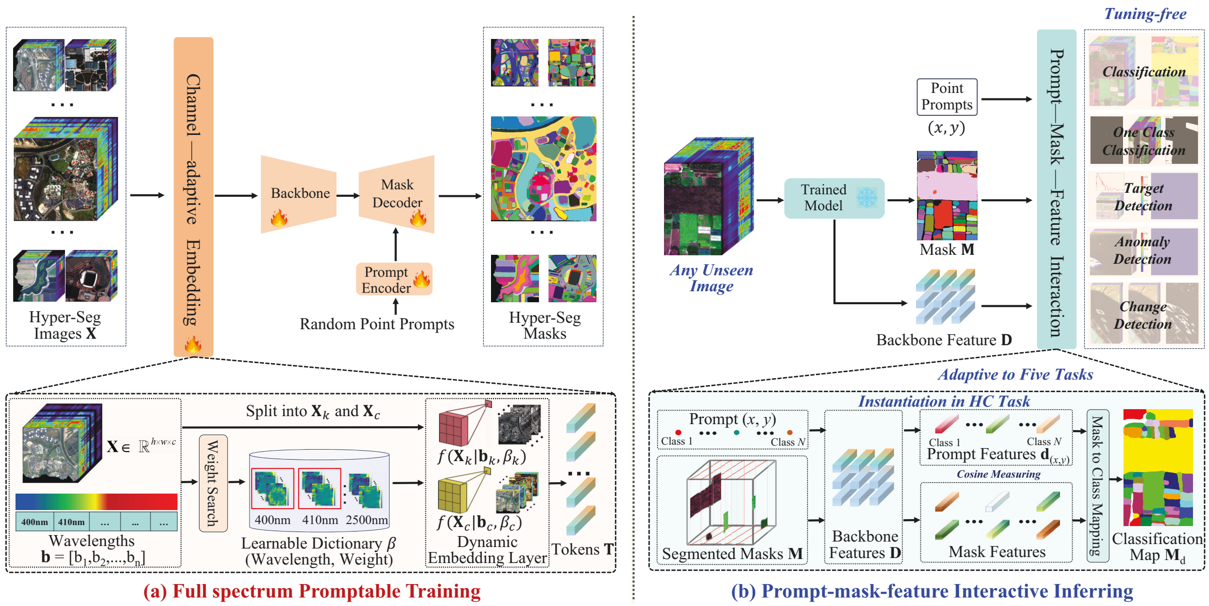


Figure 3. (a) Full spectrum Promptable Training. Based on the meta-architecture of promptable segmentation, channel-adaptive embedding layer is developed to process any-channel input, which relies on a learnable weight dictionary and generates the embedding layer dynamically according to the input wavelengths. With the constructed Hyper-Seg dataset, HyperFree is trained with randomly selected channels from full spectrum and point prompts in each iteration. (b) Prompt-mask-feature Interactive Inferring. Keep the trained HyperFree frozen, different PMF interaction workflows are designed to complete the downstream tasks in promptable or zero-shot manner, where we use feature distance to measure the semantic consistency. An interaction instantiation for classification task is given above as an example.

in Supp.2). We set two different dictionaries $\beta_{\mathbf{k}}$ and $\beta_{\mathbf{c}}$ to process $\mathbf{X_k}$ and $\mathbf{X_c}$ respectively, and the final output tokens $\mathbf{T} \in \mathbb{R}^{(h/p)\times(w/p)\times j}$ are computed as the element-wise sum, as shown in Equation (2).

$$\mathbf{T} = f(\mathbf{X_k}|\mathbf{b_k}, \beta_{\mathbf{k}}) + f(\mathbf{X_c}|\mathbf{b_c}, \beta_{\mathbf{c}}) \tag{2}$$

4.2. Full-spectrum Promptable Training

The prompt engineering paradigm is adopted to complete downstream tasks directly with designed prompts. HyperFree is thus trained in a promptable manner on constructed large-scale Hyper-Seg dataset, where the prompts are randomly chosen at each iteration and the model is expected to segment correctly the corresponding masks. Since hyperspectral image processing tasks are almost conducted at the pixel level, we choose point prompts only at the training stage. Focal loss L_f and Dice loss L_d are weighted as [17] to supervise the training as Equation (3).

$$L = 20L_f + L_d \tag{3}$$

To train the channel-adaptive embedding layer effectively, we select the subset of channels randomly from the full-spectrum range and process it with corresponding weights only in $\beta_{\mathbf{k}}$ and $\beta_{\mathbf{c}}$, which helps the learned dictionary being wavelength-aware and encode the varied channel composition into a unified token space.

4.3. Prompt-mask-feature (PMF) Interactive Inferring

After training process, the model can process unseen hyperspectral image with varied channels and segment the corresponding mask given any point prompt. However, many downstream tasks need to segment out all the masks of same semantic category rather than a single mask such as hyperspectral classification and target detection. We tackle this problem by *connecting the prompt, mask and feature together in feature space*, where prompt tells the model an object reference in task, mask outputs accurate location information, and feature reflects semantic information. In unified feature space, the cosine distance of features is used to measure the semantic similarity.

Before diving into the specific workflows, we introduce two basic interaction modes with a simple situation. Set one point (x, y) as input prompt for image $\mathbf{X}$, the corresponding feature cube $\mathbf{D}$ and all the generated masks $\mathbf{M} \in \mathbb{R}^{h\times w\times k}$ from the trained model, where $\mathbf{M}$ has k binary maps and each represents a single mask. The first interaction mode maps the prompt (x, y) into a feature vector $\mathbf{d}_{(x,y)} \in \mathbb{R}^j$ as in Equation (4), where the corresponding valid and smallest mask in $\mathbf{M}$ is used to localize the features in $\mathbf{D}$, and the mean feature is treated as $\mathbf{d}_{(x,y)}$ for the given prompt. This interaction transforms the prompt from image space to feature space, and makes it possible to explore the semantic similarity.

$$\text{Mode 1: } (x, y)|\mathbf{M}, \mathbf{D} \rightarrow \mathbf{d}_{(x,y)} \tag{4}$$

The second interaction mode selects multiple the masks in $\mathbf{M}$, which have similar semantic information with $\mathbf{d}_{(x,y)}$ as in Equation (5). The selected masks are represented as $\mathbf{M}_d$ and $\mathbf{M}_d \subseteq \mathbf{M}$. Specifically, we compute a feature representation for each single mask by localizing in $\mathbf{D}$ and computing the average, and the cosine similarity between them and $\mathbf{d}_{(x,y)}$ are treated as the semantic similarity. With some hand-designed thresholds τ, $\mathbf{M}_d$ can be finally segmented. τ represents different meanings according to the specific task.

$$\text{Mode 2: } (\mathbf{d}_{(x,y)}|\mathbf{M}, \mathbf{D}, \tau) \rightarrow \mathbf{M}_d \tag{5}$$

By using both interaction modes adaptively, five hyperspectral tasks can be completed with the trained promptable model directly. (a) **Hyperspectral classification (HC)**. Each category needs at least one prompt, and uses Mode 1 and Mode 2 in turn for each category. For a common closed-set setting, τ is not necessary since we can use the minimum feature distance between each mask representation and all the prompt representations to decide its category. Figure 3 (b) shows an exemplified workflow. (b) **Hyperspectr[al] one-class classification (HOCC)**. HOCC is similar to H[C] but the difference lies in that HOCC only needs prompts [of] the target category, and its τ is instantiated as a class pr[ior] parameter to control the mask ratio. (c) **Hyperspectral ta[r]get detection (HTD)**. Compared to HOCC, HTD acce[pts] the target spectra and it needs to be changed to a pron[mpt] by selecting the pixel with the smallest cosine distance. is instantiated as a distance threshold to convert the dens[ity] map to a binary map. (d) **Hyperspectral anomaly dete[c]tion (HAD)**. HAD is an unsupervised task and there is [no] $\mathbf{d}_{(x,y)}$ for HAD. Since HAD is very similar to unsupervis[ed] target detection, we use $\mathbf{M}$ only and filter out large obje[cts] according to mask area to output the final anomaly map. [(e)] **Hyperspectral change detection (HCD)**. Different fr[om] the prior tasks, HCD accepts two registered and temporal images to identify the changing locations. The HCD task utilizes both interaction modes for the bi-temporal images. Mode 1 processes the first image to get representation $\mathbf{d}_{(x,y)}$ for each mask, which is input to Mode 2 and compared with the corresponding mask representation with $\mathbf{M}$ and $\mathbf{D}$ of the second image. Once the temporal feature distance exceeds the preset τ, a change is considered to have happened. The interaction workflow is summarized in Table 2.

Table 2. Instantiation summary of prompts, meaning of τ, and interaction modes when HyperFree performs different tasks.

Task	Prompt Type	Hyperparameter τ	Interaction Mode
HC	One Point Per Categories	No τ	Mode 1, Mode 2
HOCC	Target Category Point	Class Prior	Mode 1, Mode 2
HTD	Target Spectra	Similarity Threshold	Mode 1, Mode 2
HAD	No Prompt	Masks Area Threshold	Mode 1
HCD	No Prompt	Similarity Threshold	Mode 1, Mode 2

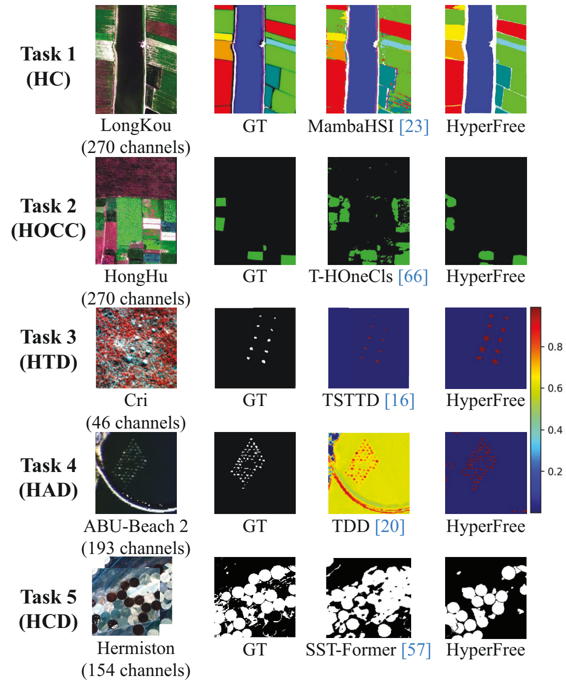


Figure 4. Exemplified comparison results for five tuning-free tasks.

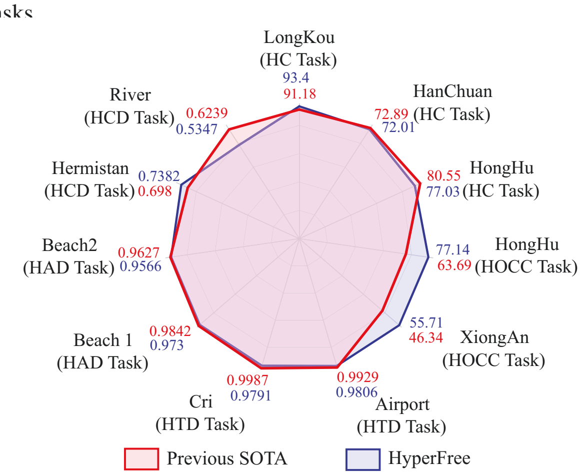


Figure 5. In tuning-free manner, HyperFree (1 prompt) can achieve comparable results with specialized models (5 shots) on 5 tasks and 11 datasets.

5. Results

5.1. Experimental Setup

We evaluate the proposed HyperFree on five downstream hyperspectral tasks including HC, HOCC, HTD, HAD and

Table 3. Quantitative comparison results on five hyperspectral tasks, where blue numbers indicates the metric ranking of HyperFree. Comparison methods were all trained on each dataset while HyperFree processed the datasets in tuning-free manner.

HC Task

Metrics	LongKou (270 channels)							HanChuan (274 channels)						
	SVM [30] (5 shot)	HybridSN [41] (5 shot)	FCN [52] (5 shot)	FPGA [68] (5 shot)	SSFTT [46] (5 shot)	MambaHSI [23] (5 shot)	HyperFree (1 prompt)	SVM [30] (5 shot)	HybridSN [41] (5 shot)	FCN [52] (5 shot)	FPGA [68] (5 shot)	SSFTT [46] (5 shot)	MambaHSI [23] (5 shot)	HyperFree (1 prompt)
OA	82.77	48.78	86.67	91.18	89.66	92.65	93.39(1st)	52.68	47.75	55.55	71.47	64.86	73.33	75.47(1st)
AA	74.02	61.37	85.60	88.35	87.96	92.57	88.03(3rd)	47.76	46.17	59.72	72.09	61.22	69.33	62.35(2nd)
KA	78.04	35.72	82.30	88.66	87.95	90.00	91.41(1st)	46.85	41.31	50.18	67.58	59.65	69.10	71.56(1st)

HOCC Task

Metrics	HongHu (270 channels)							XiongAn (256 channels)						
	OCSVM [43] (5 shot)	nnPU [18](5 shot)	BSVM [33](5 shot)	PAN [15](5 shot)	OC Loss [67](5 shot)	T-HOneCls [66](5 shot)	HyperFree (1 prompt)	OCSVM [43] (5 shot)	nnPU [18](5 shot)	BSVM [33](5 shot)	PAN [15](5 shot)	OC Loss [67](5 shot)	T-HOneCls [66](5 shot)	HyperFree (1 prompt)
F	26.33	19.13	34.82	63.69	54.73	55.97	72.52(1st)	18.31	1.76	26.30	46.34	43.08	41.34	52.50(1st)
P	56.43	19.72	50.77	80.00	58.27	46.52	68.61(2nd)	39.83	2.85	23.82	47.13	47.50	32.87	68.21(1st)
R	24.02	18.58	45.29	64.27	54.34	92.35	78.06(2nd)	16.08	1.98	57.83	53.32	47.61	60.38	65.03(1st)

HTD Task

Metrics	Airport (205 channels)							Cri (46 channels)						
	ACE [19](1 shot)	CEM [2](1 shot)	GLRT [28](1 shot)	MF [29](1 shot)	HTD-IRN [44](1 shot)	TSTTD [16](1 shot)	HyperFree (1 prompt)	ACE [19](1 shot)	CEM [2](1 shot)	GLRT [28](1 shot)	MF [29](1 shot)	HTD-IRN [44](1 shot)	TSTTD [16](1 shot)	HyperFree (1 prompt)
$AUC_{(D,F)}$	0.9794	0.9603	0.9801	0.9916	0.9745	0.9929	0.9806(3rd)	0.9735	0.9893	0.9737	0.9891	0.9975	0.9987	0.9791(5th)
AUC_{ODP}	1.5853	1.2829	1.5798	1.6968	1.4484	1.6592	1.4667(6th)	1.2015	1.4506	1.2000	1.4575	1.3995	1.6103	1.4670(2nd)

HAD Task

Metrics	Beach1 (188 channels)							Beach2 (193 channels)						
	RXD [40]	CRD [22]	ADLR [36]	LRASR [63]	Auto-AD [55]	TDD [20]	HyperFree (zero shot)	RXD [40]	CRD [22]	ADLR [36]	LRASR [63]	Auto-AD [55]	TDD [20]	HyperFree (zero shot)
$AUC_{(D,F)}$	0.9815	0.9471	0.4515	0.7461	0.9574	0.9842	0.9730(3rd)	0.9090	0.8544	0.7976	0.8225	0.9485	0.9627	0.9566(2nd)
AUC_{ODP}	1.2557	0.9785	0.5610	0.8526	1.1273	1.1383	1.9191(1st)	1.0177	0.8670	0.9064	0.8280	1.0097	1.1688	1.8697(1st)

HCD Task

Metrics	River (154 channels)							Hermiston (198 channels)						
	FC-EF [8](5 shot)	FC-Sc [8](5 shot)	FC-Sd [8](5 shot)	ML-EDAN [35](5 shot)	BIT [3](5 shot)	SST-Former [57](5 shot)	HyperFree (zero shot)	FC-EF [8](5 shot)	FC-Sc [8](5 shot)	FC-Sd [8](5 shot)	ML-EDAN [35](5 shot)	BIT [3](5 shot)	SST-Former [57](5 shot)	HyperFree (zero shot)
IoU	0.4168	0.4522	0.4534	0.3915	0.2126	0.4096	0.3649(6th)	0.3729	0.3776	0.4873	0.3252	0.5257	0.5361	0.5851(1st)
F	0.5884	0.6228	0.6239	0.5628	0.3507	0.5812	0.5347(6th)	0.5432	0.5482	0.65.52	0.4908	0.6891	0.6980	0.7382(1st)

Table 4. Compared results with SAM using the identical PMF interaction workflows.

	HC			HOCC			HTD		HAD		HCD	
	OA	AA	KA	F	Precision	Recall	$AUC_{(D,F)}$	AUC_{ODP}	$AUC_{(D,F)}$	AUC_{ODP}	F	IoU
SAM+PMF Interaction	66.62	72.99	59.41	34.13	30.35	39.47	0.7979	1.1014	0.9615	1.8844	0.5996	0.4282
HyperFree	93.40	85.66	91.42	72.52	68.61	78.06	0.9791	1.4670	0.9566	1.8697	0.7382	0.5851

HCD tasks as in Section 4.3. A total of 11 datasets were used in the evaluation (detailed in Supp.3), and each task had six comparison methods, including both classical and state-of-the-art ones. Since there is no method that can directly complete all five tasks, all comparative methods are tested after being trained on each dataset with five shots per class, while the parameters of HyperFree are fixed for all datasets with 1 prompt or zero-shot as in Table 2. We use ViT-base as backbone to conduct the experiments.

HyperFree was trained on the constructed Hyper-Seg dataset using 8 A100 GPUs for approximately 10 hours, which was based on the meta-architecture and parameters of SAM. We use AdamW optimizer with initial learning rate 8e-5 and batch size 16. To help learning the channel-adaptive embedding, we randomly select the image channels at each iteration and match it with corresponding wavelengths. 16 masks and 1 2 points per mask are randomly selected for each input image to train the promptable segmentation.

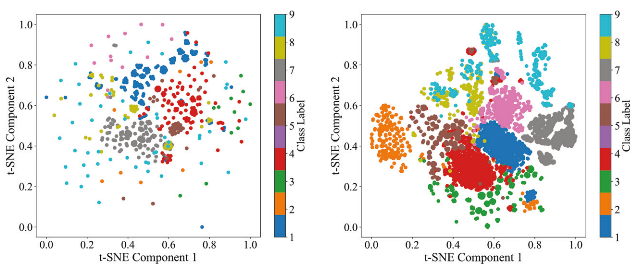
5.2. Main Results

Most tuning-free quantitative results are reported in Table 3 (1 dataset in supplement due to the space limit) and Figure 5. All the comparative methods are trained with 5 shot for each class and HyperFree processes each dataset directly without tuning. For HC, HOCC and HTD, only one point prompt is provided for each class. HAD and HCD does not need any prompt and deal with images in zero-shot manner. Table 3 shows than HyperFree can surpass most of the specialized models with frozen parameters. On both HC and HCD tasks, HyperFree even outperformed previous SOTA results, including an improvement of about 12 points in F$_1$ on HC and about 0.5 points in IoU on HCD task.

We performed qualitative comparison on five exemplified images in Figure 4, where one SOTA model have been visualized for each task comparison. Obviously, the result maps of HyperFree are more complete and smoother than the ones of specialized models. In HC, HOCC, HTD and

Table 5. Ablation results on the usage of the learned weight dictionary.

Correct Wavelengths	$\beta_\mathbf{k}$	$\beta_\mathbf{c}$	HC			HOCC			HTD		HAD		HCD	
			OA	AA	KA	F_1	Precision	Recall	$AUC_{(D,F)}$	AUC_{ODP}	$AUC_{(D,F)}$	AUC_{ODP}	F_1	IoU
✓	✓	×	73.18	64.23	65.85	65.62	47.30	64.56	0.9767	1.4644	0.9488	1.8465	0.5793	0.4078
✓	×	✓	66.63	71.52	59.44	63.62	58.61	78.44	0.9768	1.4640	0.9589	1.8768	0.7022	0.5411
×	✓	✓	75.25	38.98	66.44	57.30	64.09	54.83	0.9595	1.4273	0.9557	1.8678	0.6832	0.5188
✓	✓	✓	93.40	85.66	91.42	72.52	68.61	78.06	0.9791	1.4670	0.9566	1.8697	0.7382	0.5851



(a) HyperSigma features (b) HyperFree features

Figure 6. t-SNE visualizations of HyperSigma [54] and HyperFree features on WHU-Hi Longkou dataset. We randomly select 1000 pixels for each class.

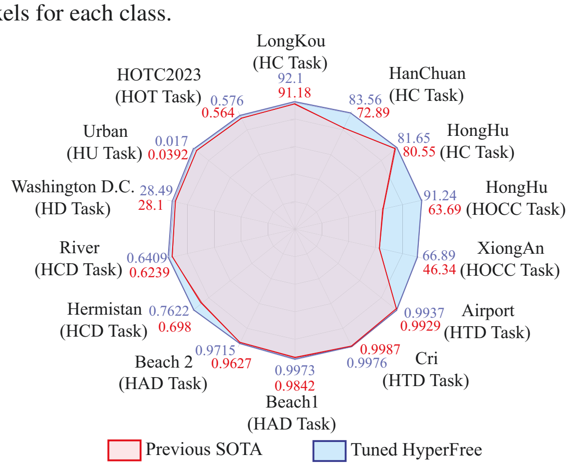


Figure 7. In tuning manner, HyperFree can achieve overall better results with specialized models on 8 tasks and 14 datasets.

HAD tasks, HyperFree can generate basically the same result as GT, which are difficulty for existing models.

5.3. Analysis Studies

HyperFree vs SAM. Proposed prompt-mask-feature interaction strategy adapts the pre-trained HyperFree to various downstream tasks. We applied the same strategy to SAM to verify the effectiveness of training on hyperspectral images and reported the results in Table 4. The metrics on most tasks have a decline of about 10∼30 points, implying a failed processing. We realize it is not a fair comparison and only use it to reflect the difference of hyperspectral data processing.

HyperFree vs HyperSigma. HyperSigma is the only open-sourced hyperspectral foundation model [54, 62], which adopts the pre-training and tuning paradigm. Thus, we cannot compare HyperFree with it in tuning-free setting directly and compare the t-SNE visualization in Figure 6(a) and Figure 6(b). In contrast, proposed HyperFree has a more compact intra-class distance obviously, supporting the tuning-free workflow of PMF interaction.

Full-Spectrum Weight Dictionary. To verify stored weights are wavelength-aware, we conducted related ablation studies in Table 5. In HC, HOCC and HCD tasks, a drop in accuracy of about 5∼40 points can be observed if we shuffled the correct wavelength order randomly, showing the wavelength-aware property of learned dictionary. Besides, we proved the built two encoding branches with dictionaries $\beta_\mathbf{k}$ and $\beta_\mathbf{c}$ can benifit each other effectively.

Tuning HyperFree. As an extensive experiment, we have also tested the tuning performance of HyperFree on 8 tasks and 14 datasets in Figure 7, which further included hyperspectral denosing (HD) [27], object tracking (HOT) [61] and unmxing (HU) tasks [37]. The tuning HyperFree has surpassed the previous SOTA models on nearly all tasks.

More analysis. For more analytical experiments, please see the supplementary material. (i) Supp.4.1: Complete qualitative comparison of 11 datasets in tuning-free manner. (ii) Supp.4.2: Complete quantitative and qualitative comparison of 14 datasets in tuning manner. (iii) Supp.4.3: Sensitivity analysis about prompt number, prompt location and the hyperparameter τ.

6. Conclusion

Hyperspectral images can provide rich spectral information and are widely used in resource monitoring and defense fields. However, its unique properties increase the difficulty of foundation model development such as varied-length channels, high sensitivity to imaging conditions and the acquirement difficulty, making the mainstream P-T being low cost-effectiveness due to the image-by-image tuning. Our work tackles this problem by converting to adapt the visual P-E [17]. Differently, we design a learnable weight dictionary covering full-spectrum to generate the channel-adaptive embeddings and PMF interaction to generate semantic-aware masks for different tasks. We hope HyperFree can speed up the era of "GPT moment" for hyperspectral intelligent processing.

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